

1.2 Deep Neural Networks

Artificial Neurons & Neural Networks

The idea behind artificial neural networks is to mimic the functionality of the human brain and can be traced back to MCCULLOCH and PITTS in 1943, who were interested in developing an algorithmic approach to learning.

DEFINITION 1.2.1 (ARTIFICIAL NEURON)

An **artificial neuron** with **weights** $w \in \mathbb{R}^d$, **bias** $b \in \mathbb{R}$ and **activation function** $\varrho : \mathbb{R} \rightarrow \mathbb{R}$ is a function

$$\mathbb{R}^n \rightarrow \mathbb{R}, x \mapsto \varrho(\langle x, w \rangle - b).$$

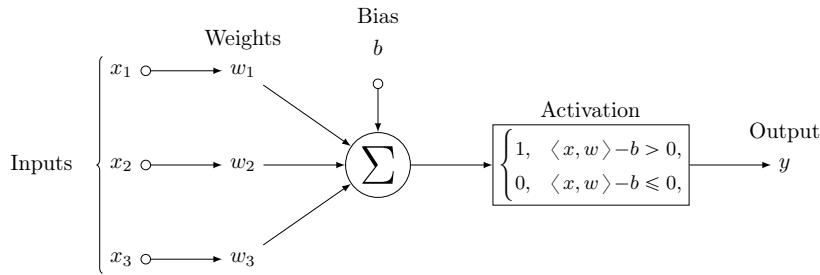


Fig. 1: An artificial neuron.

Example 1.2.2 (Activation functions)

- HEAVISIDE function $\mathbb{1}_{[0, \infty)}$,
- Sigmoid function $x \mapsto (1 + e^{-x})^{-1}$,
- Rectifiable Linear Unit (ReLU) $x \mapsto \max(0, x)$,
- Softmax function $x \mapsto \ln(1 + e^x)$.

◇

artificial neuron
activation function

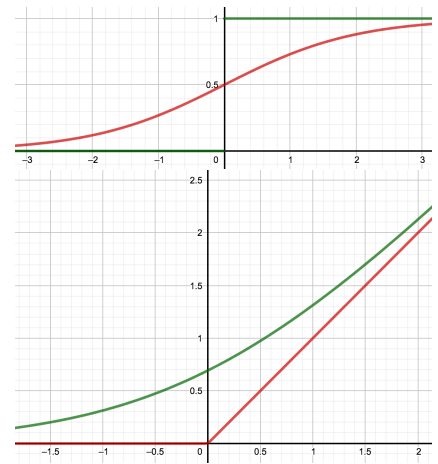


Fig. 2: Activation functions. [1]
artificial neural network

DEFINITION 1.2.3 (ARTIFICIAL NEURAL NETWORK TYPES)

An **artificial neural network** (ANN) is a **graph** consisting of **artificial neurons** called feed-forward ANN if it is direct and acyclic and recurrent ANN else.

Deep Neural Networks

The start of the **Age of Data** at the turn of the century saw a drastic improvement of computing power and vast amounts of data being available. This gave rise to deep neural networks (cf. Y. LECUN et al., ca 2000).

DEFINITION 1.2.4 ((DEEP) NEURAL NETWORK (DNN))

The map

$$\mathbb{R}^d \rightarrow \mathbb{R}^{N_L}, x \mapsto T_L \varrho(T_{L-1} \varrho(\dots \varrho(T_1(x))))$$

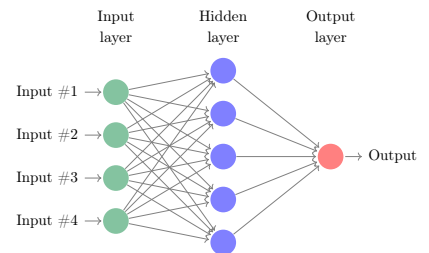


Fig. 3: An artificial neural network.

is called DNN, where $d := N_0 \in \mathbb{N}$ is the dimension of input layer, L the number of layers, N_ℓ the number of neurons in layer $\ell \in [L]$, $\varrho : \mathbb{R} \rightarrow \mathbb{R}$ the activation function and $(T_\ell : \mathbb{R}^{N_{\ell-1}} \rightarrow \mathbb{R}^{N_\ell})_{\ell=1}^L$ are affine linear maps.

Example 1.2.5 (DNN Architecture: Affine linear maps)

The affine linear maps T_ℓ are defined by a matrix $A_\ell \in \mathbb{R}^{N_{\ell-1} \times N_\ell}$ and an affine part $b_\ell \in \mathbb{R}^{N_\ell}$ via $T_\ell(x) := A_\ell x + b_\ell$. $\diamond$

In the example on the right

$$A_1 := \begin{pmatrix} a_1^1 & a_2^1 & 0 \\ 0 & 0 & a_3^1 \\ 0 & 0 & a_4^1 \end{pmatrix} \quad \text{and} \quad A_2 := \begin{pmatrix} a_1^2 & a_2^2 & 0 \\ 0 & 0 & a_3^2 \end{pmatrix}.$$

Example 1.2.6 (Training of neural networks)

Let $\mathcal{M}$ be a manifold and $(x_i, f(x_i))_{i=1}^m$ samples of a function $f : \mathcal{M} \rightarrow [K]$, aiming to **classify** an input $x \in \mathcal{M}$ into one of K classes.

We now select $d, L, (N_\ell)_{\ell=1}^L$ and ϱ , collectively forming the **architecture** of a DNN. Now we **learn the affine linear functions** $(T_\ell)_{\ell=1}^L$ by

$$\min_{(A_\ell, b_\ell)_{\ell=1}^L} \sum_{i=1}^m \mathcal{L}(\Phi_{(A_\ell, b_\ell)}(x_i), f(x_i)) + \lambda \mathcal{R}((A_\ell, b_\ell)_{\ell=1}^L).$$

where $\mathcal{L}$ is a loss function, $\mathcal{R}$ a regularization term and $\lambda > 0$ (cf. 1.2.9) yielding the network

$$\Phi_{(A_\ell, b_\ell)_\ell} : \mathbb{R}^d \rightarrow \mathbb{R}^{N_L}, \quad x \mapsto T_L \varrho(T_{L-1} \varrho(\dots \varrho(T_1(x))))$$

This is often solved by **stochastic gradient descent** (cf. subsection 1.3), in pursuit of

$$\Phi_{(A_\ell, b_\ell)_\ell} \approx f. \quad \diamond$$

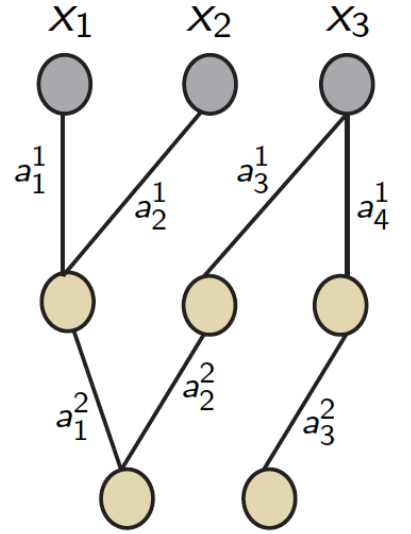


Fig. 4: Neural network architecture. [2]



Four Fundamental Questions concerning DNNs

- **Expressivity** ($\rightsquigarrow$ Applied Harmonic Analysis, Approximation Theory)
 - How powerful is the network architecture?
 - Can it represent the correct functions?
- **Learning** ($\rightsquigarrow$ Differential Geometry, Optimal Control, Optimization)
 - Why does the algorithm yield reasonable results?
 - What are sensible starting values?
- **Generalization** ($\rightsquigarrow$ Learning Theory, Optimization, Statistics)
 - Why do DNNs preform well on unseen data?
 - What impact does the depth of the network have?
- **Interpretability** ($\rightsquigarrow$ Information Theory, Uncertainty Quantification)

Expressivity

Learning

Generalization

Interpretability